Final exam · Unit 3 · Constant coefficients, Cauchy-Euler and forcing · 2.5 points · 50 min · High difficulty / trap

P16 — x^2 y'' - x y' + 2y = x sin(ln x): reduction to constant coefficients by x = e^z, a resonant particular solution x ln x cos(ln x), and a boundary-value problem on [1, b] whose existence and uniqueness depend on b

Consider, for $x \geq 1$,

$$x^2 y'' - x\,y' + 2y = x\sin(\ln x).$$

(a) With the change of variable $x = e^z$, $Y(z) = y(e^z)$, write the chain rule for y' and y'' explicitly and obtain a constant-coefficient equation for Y. Find the general solution of the homogeneous equation in terms of x.

(b) Find a particular solution. Explain why the natural guess $x\big(A\cos\ln x + B\sin\ln x\big)$ fails.

(c) Consider the boundary-value problem formed by the equation and $y(1) = 0$, $y(b) = 0$, with $b > 1$.

- *For which b does it have exactly one solution? Write it, and simplify it for $b = e^{\pi/2}$.*
- *Show that for $b = e^{n\pi}$ ($n = 1, 2, \ldots$) it has no solution.*
- *For $b = e^{\pi}$, replace $y(b) = 0$ by $y(e^{\pi}) = \beta$. For which β are there solutions, and how many?*

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknown $y(x)$ on $[1, b]$; $x > 0$ so $\ln x$ is defined and the leading coefficient x^2 does not vanish.
- Operator: linear, second order, variable coefficients of Cauchy-Euler type ($x^k y^{(k)}$), non-homogeneous with forcing $x\sin(\ln x)$.
- Two conditions at **different** points: a boundary-value problem, for which Picard's theorem gives no existence or uniqueness guarantee.

Phase 2. Canonical formulation

Standard form: $y'' - \dfrac{1}{x}\,y' + \dfrac{2}{x^2}\,y = \dfrac{\sin(\ln x)}{x}$. After $x = e^z$ the problem becomes a constant-coefficient equation on $0 \leq z \leq \ln b$ with $Y(0) = 0, Y(\ln b) = 0$.

Part (a) · 0.75 points

Change of variable and homogeneous solution

Step 1. First derivative

$y(x) = Y(z(x))$ with $z = \ln x$. Chain rule: $y' = \dfrac{dY}{dz}\dfrac{dz}{dx} = \dfrac{1}{x}\,Y'(z)$.

Step 2. Second derivative

Product rule on $\dfrac{1}{x} \cdot Y'(z(x))$, and chain rule on the second factor, $\dfrac{d}{dx}Y'(z) = Y''(z)\dfrac{1}{x}$:

$$y'' = -\frac{1}{x^2}Y' + \frac{1}{x}\cdot\frac{1}{x}Y'' = \frac{1}{x^2}\big(Y'' - Y'\big)$$

Step 3. Transformed equation

$x^2y'' = Y'' - Y',\ xy' = Y',\ x = e^z,\ \sin(\ln x) = \sin z$:

$$Y'' - Y' - Y' + 2Y = e^z \sin z \implies Y'' - 2Y' + 2Y = e^z \sin z$$

Step 4. Homogeneous solution

Characteristic polynomial $r^2 - 2r + 2 = 0, r = \dfrac{2 \pm \sqrt{4-8}}{2} = 1 \pm i$. So $Y_h = e^z(C_1\cos z + C_2 \sin z)$ and, back with $e^z = x$, $z = \ln x$:

$$y_h = x\big(C_1\cos(\ln x) + C_2\sin(\ln x)\big)$$

KEY RESULTS

- $y' = \dfrac{1}{x}\dfrac{dY}{dz},\ y'' = \dfrac{1}{x^2}\Big(\dfrac{d^2Y}{dz^2} - \dfrac{dY}{dz}\Big)$.
- $Y'' - 2Y' + 2Y = e^z \sin z$; characteristic roots $r = 1 \pm i$.
- $y_h = x\big(C_1\cos(\ln x) + C_2\sin(\ln x)\big)$.

Part (b) · 0.75 points

Resonant particular solution

Step 1. Why the natural guess fails

$\alpha + i\beta = 1 + i$ is a simple root of $r^2 - 2r + 2$; $e^z \cos z$ and $e^z \sin z$ solve the homogeneous equation, so any $e^z(A\cos z + B\sin z)$ is mapped to 0 and cannot produce $e^z \sin z$. The general rule (Robinson Sec. 14.5; Notes.pdf, the second case of the guess) multiplies the guess by z once (simple root).

Step 2. Shift by e^z

Write $Y = e^z w(z)$. Product rule: $Y' = e^z(w' + w), Y'' = e^z(w'' + 2w' + w)$. Then

$$Y'' - 2Y' + 2Y = e^z\big[w'' + 2w' + w - 2w' - 2w + 2w\big] = e^z\big(w'' + w\big)$$

so the equation becomes $w'' + w = \sin z$, and the guess is $w = z(A\cos z + B\sin z)$.

Step 3. Coefficients

$w' = A\cos z + B\sin z + z(-A\sin z + B\cos z)$ and

$$w'' = 2(-A\sin z + B\cos z) - z(A\cos z + B\sin z)$$

Hence $w'' + w = -2A\sin z + 2B\cos z = \sin z$: $-2A = 1, 2B = 0$, so $A = -\tfrac{1}{2}, B = 0$.

Step 4. Back to x

$Y_p = -\tfrac{1}{2} z e^z \cos z$, and with $z = \ln x$, $e^z = x$:

$$y_p = -\frac{1}{2}\, x \ln x\, \cos(\ln x)$$

the $x^r \ln x$ structure typical of resonance in Euler equations (here $r = 1 \pm i$). General solution:

$$y = x\big(C_1\cos(\ln x) + C_2\sin(\ln x)\big) - \frac{1}{2} x \ln x \cos(\ln x)$$

KEY RESULTS

- The guess $e^z(A\cos z + B\sin z)$ is a homogeneous solution: substituting gives $0 = e^z \sin z$, impossible.
- Corrected guess $Y_p = ze^z(A\cos z + B\sin z)$; $A = -\frac{1}{2}$, $B = 0$.
- $y_p = -\frac{1}{2}\, x \ln x \cos(\ln x)$; general solution $y = x\big(C_1 \cos\ln x + C_2 \sin\ln x\big) - \frac{1}{2} x \ln x \cos\ln x$.

Part (c) · 1 point

Boundary-value problem on [1, b]

Step 1. Condition at x = 1

$\ln 1 = 0, \cos 0 = 1, \sin 0 = 0$: $y(1) = C_1$. So $C_1 = 0$ and $y = C_2 x \sin(\ln x) - \frac{1}{2} x \ln x \cos(\ln x)$.

Step 2. Condition at x = b

$y(b) = b\Big(C_2 \sin(\ln b) - \frac{1}{2}\ln b \cos(\ln b)\Big) = 0$, and $b > 0$, so

$$C_2 \sin(\ln b) = \tfrac{1}{2} \ln b \cos(\ln b)$$

Step 3. Unique case

If $\sin(\ln b) \neq 0$, i.e. $b \neq e^{n\pi}$, then $C_2 = \frac{1}{2}\ln b \cot(\ln b)$ and

$$y = \frac{x}{2}\Big(\ln b\, \cot(\ln b) \sin(\ln x) - \ln x \cos(\ln x)\Big)$$

For $b = e^{\pi/2}$: $\cot(\pi/2) = 0$, so $C_2 = 0$ and the solution is y_p itself, which already vanishes at $x = 1$ and at $x = e^{\pi/2}$ (where $\cos(\ln x) = 0$).

Step 4. b = e^{n pi}: no solution

$\sin(n\pi) = 0$ and $\cos(n\pi) = (-1)^n$, so Step 2 reads $0 = \frac{1}{2}\, n\pi (-1)^n$, impossible. Equivalently $y(e^{n\pi}) = (-1)^{n+1}\frac{n\pi}{2} e^{n\pi}$ whatever C_2 is. The reason: $x\sin(\ln x)$ is a non-trivial solution of the **homogeneous** boundary-value problem (in z, $\sin z$ vanishes at $z = 0$ and $z = n\pi$, exactly as the eigenfunctions of Haberman Sec. 2.3), and the forcing is not orthogonal to it. Fredholm alternative (extension): the non-homogeneous problem is solvable if and only if the forcing, written for the self-adjoint form of the operator, is orthogonal to every solution of the homogeneous problem. Dividing the equation by x^3 gives the self-adjoint form $\left(\frac{y'}{x}\right)' + \frac{2}{x^3} y = \frac{\sin(\ln x)}{x^2}$, and with $z = \ln x$, $dz = dx/x$:

$$\int_1^{e^{n\pi}} \frac{\sin(\ln x)}{x^2}\, x\sin(\ln x)\, dx = \int_0^{n\pi} \sin^2 z\, dz = \Big[\frac{z}{2} - \frac{\sin 2z}{4}\Big]_0^{n\pi} = \frac{n\pi}{2} \neq 0$$

Step 5. Changed data at b = e^pi

With $y(e^\pi) = \beta$: $e^\pi\big(C_2 \cdot 0 - \frac{1}{2}\pi(-1)\big) = \beta$, i.e. $\beta = \dfrac{\pi e^\pi}{2} \approx 36.35$. If β has this value, C_2 is free: infinitely many solutions $y = y_p + C_2 x\sin(\ln x)$. For any other β, none.

KEY RESULTS

- $y(1) = 0 \Rightarrow C_1 = 0$. $y(b) = 0 \Rightarrow C_2 \sin(\ln b) = \frac{1}{2}\ln b \cos(\ln b)$.
- Unique solution iff $b \neq e^{n\pi}$: $y = \dfrac{x}{2}\Big(\ln b\cot(\ln b)\sin(\ln x) - \ln x\cos(\ln x)\Big)$. For $b = e^{\pi/2}$: $y = y_p = -\frac{1}{2}x\ln x\cos(\ln x)$.
- $b = e^{n\pi}$: the condition reads $0 = (-1)^{n+1}\dfrac{n\pi}{2}e^{n\pi} \neq 0$: no solution.
- $y(e^\pi) = \beta$: solvable iff $\beta = \dfrac{\pi e^\pi}{2} \approx 36.35$, and then there are infinitely many solutions $y = y_p + C_2 x\sin(\ln x)$.

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** $b \to e^{n\pi}$ from either side: $\cot(\ln b) \to \pm\infty$, so the unique solution blows up, the signature of an approaching eigenvalue. $b \to 1^+$: $\ln b\cot(\ln b) \to 1$ and $y \to \frac{x}{2}\big(\sin(\ln x) - \ln x\cos(\ln x)\big)$ restricted to a shrinking interval, which tends to 0.
- **Checks:** y_p substituted back in the original variable: the script verifies $x^2y_p'' - xy_p' + 2y_p = x\sin(\ln x)$; the boundary values $y(1) = 0$, $y(b) = 0$ hold identically in b; the solution of Step 3 is the case $b = e$ with $C_2 = \frac{1}{2}\cot 1 \approx 0.32105$.
- **Common mistakes in the exam:** (1) writing $y'' = \dfrac{1}{x^2}Y''$, forgetting the $-Y'$ term produced by the product rule; (2) using the non-resonant guess and getting the contradiction $0 = e^z\sin z$ without multiplying by z; (3) concluding "unique by Picard" for a boundary-value problem; (4) assuming that a boundary-value problem with a zero-determinant system always has infinitely many solutions: it has none unless the data satisfy the solvability condition.